

Evidence for novel epigenetic marks within plants.

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Variation in patterns of gene expression can result from modifications in the genome that occur without a change in the sequence of the DNA; such modifications include methylation of cytosine to generate 5-methylcytosine (5mC) resulting in the generation of heritable epimutation and novel epialleles. This type of non-sequence variation is called epigenetics. The enzymes responsible for generation of such DNA modifications in mammals are named DNA methyltransferases (DNMT) including DNMT1, DNMT2 and DNMT3. The later stages of oxidations to these modifications are catalyzed by Ten Eleven Translocation (TET) proteins, which contain catalytic domains belonging to the 2-oxoglutarate dependent dioxygenase family. In various mammalian cells/tissues including embryonic stem cells, cancer cells and brain tissues, it has been confirmed that these proteins are able to induce the stepwise oxidization of 5-methyl cytosine to 5-hydroxymethylcytosine (5hmC), 5-formylcytosine (5fC), and finally 5-carboxylcytosine (5caC). Each stage from initial methylation until the end of the DNA demethylation process is considered as a specific epigenetic mark that may regulate gene expression. This review discusses controversial evidence for the presence of such oxidative products, particularly 5hmC, in various plant species. Whereas some reports suggest no evidence for enzymatic DNA demethylation, other reports suggest that the presence of oxidative products is followed by the active demethylation and indicate the contribution of possible TET-like proteins in the regulation of gene expression in plants. The review also summarizes the results obtained by expressing the human TET conserved catalytic domain in transgenic plants.

Profiles of a broad spectrum of epigenetic DNA modifications in normal and malignant human cell lines: Proliferation rate is not the major factor responsible for the 5-hydroxymethyl-2'-deoxycytidine level in cultured cancerous cell lines.

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Active demethylation of 5-methylcytosine moiety in DNA occurs by its sequential oxidation to 5-hydroxymethylcytosine, 5-formylcytosine and 5-carboxycytosine, catalysed by enzymes of the Ten-Eleven Translocation family proteins (TETs 1, 2 and 3). Here we analyzed for the first time all the intermediate products of DNA demethylation pathway in the form of deoxynucleosides (5-methyl-2'-deoxycytidine, 5-(hydroxymethyl)-2'-deoxycytidine, 5-formyl-2'-deoxycytidine and 5-carboxy-2'-deoxycytidine as well as 5-(hydroxymethyl)-2'-deoxyuridine) using automated isotope-dilution online two-dimensional ultra-performance liquid chromatography with tandem mass spectrometry. DNA was isolated from human malignant cell lines of colon adenocarcinoma (HCT 116), melanoma (Me45), myelogenous leukemia bone marrow blasts (K562), EBV-positive Burkitt's lymphoma lymphoblasts (Raji), EBV-negative Burkitt's lymphoma lymphoblasts (male-CA46 and female-ST486), as well as normal neonatal dermal fibroblasts (NHDF-Neo). The expression levels of TET1, TET2, TET3, SMUG1, and TDG genes were also assayed by RT-qPCR. Our results show a global erasure of 5-hydroxymethyl-2'-deoxycytidine and 5-carboxy-2'-deoxycytidine in DNA of cultured cells compared with DNA from primary malignant tissue. Moreover, malignant cells in culture have a quite different DNA epigenetic profile than cultured normal cells, and different types of malignant cells display different and characteristic profiles of DNA epigenetic marks. Similar analyses of a broader spectrum of epigenetic modifications, not restricted to 5-methyl-2'-deoxycytidine, could lead to better understanding of the mechanism(s) responsible for emergence of different types of cancer cells.

Enzymatic DNA oxidation: mechanisms and biological significance.

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DNA methylation at cytosines (5mC) is a major epigenetic modification involved in the regulation of multiple biological processes in mammals. How methylation is reversed was until recently poorly understood. The family of dioxygenases commonly known as Ten-eleven translocation (Tet) proteins are responsible for the oxidation of 5mC into three new forms, 5-hydroxymethylcytosine (5hmC), 5-formylcytosine (5fC) and 5-carboxylcytosine (5caC). Current models link Tet-mediated 5mC oxidation with active DNA demethylation. The higher oxidation products (5fC and 5caC) are recognized and excised by the DNA glycosylase TDG via the base excision repair pathway. Like DNA methyltransferases, Tet enzymes are important for embryonic development. We will examine the mechanism and biological significance of Tet-mediated 5mC oxidation in the context of pronuclear DNA demethylation in mouse early embryos. In contrast to its role in active demethylation in the germ cells and early embryo, a number of lines of evidence suggest that the intragenic 5hmC present in brain may act as a stable mark instead. This short review explores mechanistic aspects of TET oxidation activity, the impact Tet enzymes have on epigenome organization and their contribution to the regulation of early embryonic and neuronal development.

TET1 overexpression affects cell proliferation and apoptosis in aging ovaries.

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Along with the progress of society, human life expectancy has been increasing, and late marriage and late childbearing are the current trend. Since reproductive aging affects fertility, ovarian aging in women has become a major reproductive health issue in the current society. During ovarian aging, DNA methylation levels may change. The ten-eleven translocation (TET) protein family proteins TET1, TET2, and TET3 are important DNA demethylation enzymes, and differential expression of TET1, TET2, and TET3 may affect the proliferation and apoptosis of aging ovarian cells. The aim of this study was to investigate the role of TET1 in the regulation of ovarian aging. The expression of 5-methylcytosine (5mC) and 5-hydroxymethylcytosine (5hmC) was analyzed by immunofluorescence (IF) in young and aging ovaries of six 6-8-week-old female mice and six 6-8-month-old female mice. Then, the expression pattern of the TET protein family in young and aging ovaries of mice was investigated. To determine the impact of TET1 on ovarian development, the aging of IOSE-80, KGN, and SKOV-3 cells was induced with D-galactosidase (D-gal). Cells were then transfected using the TET1 overexpression vector or si-TET1. We assessed the proliferation and apoptosis of aging cells after transfection and analyzed the regulatory effect of TET1 expression on aging cells. Additionally, we verified the Tet1 expression in Tet1-KO mice. The 5mC to 5hmC transition, oocyte maturation, and blastocyst rate were reduced in aging mice compared to young mice. In aging mice ovaries, the expression levels of Tet1, Tet2, and Tet3 were reduced significantly, with Tet1 being particularly pronounced. The overexpression of TET1 promoted proliferation and inhibited apoptosis in aging human ovarian cells. Furthermore, Tet1 expression was very low in Tet1-KO C57BL/6 J mice ovaries. This study demonstrates that the expression levels of TET family proteins are low in aging ovaries, and the overexpression of TET1 can promote proliferation and inhibit apoptosis in aging ovarian cells.

Tet2-mediated DNA demethylation regulates the proliferation and apoptosis of human leukemia K562 cells.

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TET2 is a member of the TET protein family which is responsible for active DNA demethylation through catalyzing the successive oxidation of 5-methylcytosine (5mC) to 5-hydroxymethylcytosine (5hmC), 5-formylcytosine (5fC), and 5-carboxylcytosine (5caC), and mutations of Tet2 frequently lead to hematological malignancies. However, the relationship between Tet2-mediated demethylation and hematological malignancies is unclear. The human leukemia K562 cell line is an immortalized leukemia line that serves as an in vitro model of erythroleukemia. In this study, we investigated the effect of Tet2-mediated demethylation on the apoptosis and proliferation of human leukemia K562 cells and found that knockdown of Tet2 promoted and inhibited K562 cell proliferation and apoptosis, respectively, while upregulation of TET2 enzymatic activity via alpha-ketoglutaric acid (α -KG) had the opposite effects. Therefore, the Tet2 gene acts as a potential target for the treatment of leukemia, and small molecules that target the Tet2 gene may be used to screen antitumor drugs for hematological malignancies.

Inhibition of TET-mediated DNA demethylation suppresses osteoblast differentiation.

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DNA methylation is an epigenetic modification critical for the regulation of chromatin structure and gene expression during development and disease. The ten-eleven translocation (TET) enzyme family catalyzes the hydroxymethylation and subsequent demethylation of DNA by oxidizing 5-methylcytosine (5mC) to 5-hydroxymethylcytosine (5hmC). Little is known about TET protein function due to a lack of pharmacological tools to manipulate DNA hydroxymethylation levels. In this study, we examined the role of TET-mediated DNA hydroxymethylation during BMP-induced C2C12 osteoblast differentiation using a novel cytosine-based selective TET enzyme inhibitor, Bobcat339 (BC339). Treatment of C2C12 cells with BC339 increased global 5mC and decreased global 5hmC without adversely affecting cell viability, proliferation, or apoptosis. Furthermore, BC339 treatment inhibited osteoblast marker gene expression and decreased alkaline phosphatase activity during differentiation. Methylated DNA immunoprecipitation and bisulfite sequencing showed that inhibition of TET with BC339 led to increased 5mC at specific CpG-rich regions at the promoter of Sp7, a key osteoblast transcription factor. Consistent with promoter 5mC marks being associated with transcriptional repression, luciferase activity of an Sp7-promoter-reporter construct was repressed by in vitro DNA methylation or BC339. Chromatin immunoprecipitation analysis confirmed that TET2 does indeed occupy the promoter region of Sp7. Accordingly, forced overexpression of SP7 rescued the inhibition of osteogenic differentiation by BC339. In conclusion, our data suggest that TET-mediated DNA demethylation of genomic regions, including the Sp7 promoter, plays a role in the initiation of osteoblast differentiation. Furthermore, BC339 is a novel pharmacological tool for the modulation of DNA methylation dynamics for research and therapeutic applications.

PROSER1 mediates TET2 O-GlcNAcylation to regulate DNA demethylation on UTX-dependent enhancers and CpG islands.

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DNA methylation at enhancers and CpG islands usually leads to gene repression, which is counteracted by DNA demethylation through the TET protein family. However, how TET enzymes are recruited and regulated at these genomic loci is not fully understood. Here, we identify TET2, the glycosyltransferase OGT and a previously undescribed proline and serine rich protein, PROSER1 as interactors of UTX, a component of the enhancer-associated MLL3/4 complexes. We find that PROSER1 mediates the interaction between OGT and TET2, thus promoting TET2 O-GlcNAcylation and protein stability. In addition, PROSER1, UTX, TET1/2, and OGT colocalize on many genomic elements genome-wide. Loss of PROSER1 results in lower enrichment of UTX, TET1/2, and OGT at enhancers and CpG islands, with a concomitant increase in DNA methylation and transcriptional down-regulation of associated target genes and increased DNA hypermethylation encroachment at H3K4me1-predisposed CpG islands. Furthermore, we provide evidence that PROSER1 acts as a more general regulator of OGT activity by controlling O-GlcNAcylation of multiple other chromatin signaling pathways. Taken together, this study describes for the first time a regulator of TET2 O-GlcNAcylation and its implications in mediating DNA demethylation at UTX-dependent enhancers and CpG islands and supports an important role for PROSER1 in regulating the function of various chromatin-associated proteins via OGT-mediated O-GlcNAcylation.

Ten-eleven translocation 2 modulates allergic inflammation by 5-hydroxymethylcytosine remodeling of immunologic pathways.

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Allergic rhinitis (AR) is an allergen-specific immunoglobulin E-mediated inflammatory disease. Both genetic and environmental factors could play a role in the pathophysiology of AR. 5-methylcytosine (5mC) can be converted to 5-hydroxymethylcytosine (5hmC) by the ten-eleven translocation (Tet) family of proteins as part of active deoxyribonucleic acid (DNA) demethylation pathway. 5hmC plays an important role in the regulation of gene expression and differentiation in immune cells. Here, we show that loss of Tet protein 2 (Tet2) could impact the severity of AR in the ovalbumin-induced mouse model. Genome-wide 5hmC profiling of both wild-type and Tet2 KO mice in response to AR revealed that the loss of Tet2 could lead to 5hmC alteration at specific immune response genes. Both partial loss and complete loss of Tet2 alters the 5hmC dynamic remodeling for the adaptive immune pathway as well as cytokines. Thus, our results reveal a new role of Tet2 in immunology, and Tet2 may serve as a promising target in regulating the level of immune response.

TET3 dioxygenase modulates gene conversion at the avian immunoglobulin variable region via demethylation of non-CpG sites in pseudogene templates.

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Diversification of the avian primary immunoglobulin (Ig) repertoire is achieved in developing B cells by somatic hypermutation (SHM) and gene conversion (GCV). GCV is a type of homologous recombination that unidirectionally transfers segments of Ig pseudogenes to Ig variable domains. It is regulated by epigenetic mechanisms like histone modifications, but the role of DNA methylation remains unclear. Here, we demonstrate that the chicken B-cell line DT40 lacking TET3, a member of the TET (Ten-eleven translocation) family dioxygenases that facilitate DNA demethylation, exhibited a marked reduction in

GCV activity in Ig variable regions. This was accompanied by a drop in the bulk levels of 5-hydroxymethylcytosine, an oxidized derivative of 5-methylcytosine, whereas TET1-deficient or TET2-deficient DT40 strains did not exhibit such effects. Deletion of TET3 caused little effects on the expression of proteins required for SHM and GCV, but induced hypermethylation in some Ig pseudogene templates. Notably, the enhanced methylation occurred preferably on non-CpG cytosines. Disruption of both TET1 and TET3 significantly inhibited the expression of activation-induced cytidine deaminase (AID), an essential player in Ig diversification. These results uncover unique roles of TET proteins in avian Ig diversification, highlighting the potential importance of TET3 in maintaining hypomethylation in Ig pseudogenes.

Quantification of global DNA methylation level using 5-methylcytosine dioxygenase.

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DNA methylation is one of the best studied epigenetic modifications. Alteration of the global DNA methylation level occurs in abnormal cells, such as those associated with cancers and Alzheimer's disease. Several assays are used to determine the global DNA methylation level, including the bisulfite-based assay, high-performance liquid chromatography (HPLC)-based assay, enzyme-linked immunosorbent assay (ELISA), and methyl acceptance assay. However, these assays require several cumbersome steps to detect methylation levels. We developed a simpler enzymatic assay for the quantification of the global DNA methylation level using the Ten-eleven translocation (TET) protein. TET proteins mediate DNA demethylation through the oxidation of 5-methylcytosine (5mC) in CpG in mammalian cells. Succinate is produced during this oxidation reaction, and the amount of succinate produced correlates to the global DNA methylation level. The catalytic domain of the TET2 was expressed in *Escherichia coli* (*E. coli*), and the purified TET2 catalytic domain was reacted with human genomic DNA. The reaction solution was used for enzymatic succinate quantification with no purification step. The results showed that the succinate produced through TET-mediated oxidation increased with increasing global DNA methylation levels in human genomic DNA, which was determined using the bisulfite method. These results show that the global DNA methylation level is quantifiable by measuring the amount of succinate produced by the TET2-mediated 5mC oxidation reaction. Graphical abstract.